

Course: Information Visualization, Data Science and Social Media Analytics (Module 2)

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Data repository: https://github.com/AimBtlv/research-NetworkCitation_Eddubba.git (/ 8-output_Modul2)

Visualising Bibliographic Citation Networks in Edubba Studies

Abstract

This project analyses how scholarly knowledge about Edubba schools of the Old Babylonian period is structurally organised within academic citation networks. Using a corpus of 2,103 publications, the study applies network analysis and information visualisation to examine how primary data, mediators, and authoritative interpretations are connected through citation relationships. Five hypotheses concerning network topology and reliability were tested through a coordinated visual analytics pipeline combining statistical charts and node-link network visualisations. The project approaches these questions through information visualisation and network analysis, using visual encodings to reveal structural patterns not observable through linear bibliographic reading. The project demonstrates that visual analytics can reveal hidden epistemic structures in humanities research.

Introduction

The study of Edubba schools in the Old Babylonian period (20th–17th centuries BCE) has produced a large and fragmented body of scholarship. While archaeological evidence in the form of educational tablets is abundant, interpretations of their institutional context remain indirect and often reconstructed through secondary literature. This project does not aim to reinterpret Edubba history directly. Instead, it treats scholarly publications as a citation network in order to analyse how knowledge about Edubba is structurally produced, transmitted, and stabilised within academic discourse. The focus is therefore not on what is known about Edubba, but on how that knowledge is organised across the scholarly ecosystem. To make these structures visible, the study combines network analysis with a systematic visualisation pipeline. Citation relations are modelled as directed edges between publications, enabling the construction of a multi-layered epistemic network. Through a consistent visual encoding system (node colour, layout, size, and centrality-based mappings), the figures reveal patterns that are not accessible through linear reading or bibliographic inspection alone. The guiding assumption is that epistemic structure becomes observable only when translated into visual form, where relational properties such as mediation, centrality, and fragmentation can be directly perceived.

Research Questions and Hypotheses

RQ1 - How is scholarly knowledge about Edubba schools structurally organised within the academic citation network and what role do primary archaeological sources play within this structure?

RQ2 - Which sources can be identified as epistemically reliable based on their structural position relative to primary data and do the most reliable sources coincide with the most cited ones?

To assess the reliability and epistemic role of sources in the corpus, this study employs a custom Step-Distance Model.¹ Full operational definitions and assignment criteria of Step-Distance Model are provided in the Methods section and in description of the structure of the method.² Due to which it became possible to carry out a quantitative analysis of several hypotheses. Each hypothesis emerges from the result of the previous one, forming a chain of evidence rather than a set of independent tests.

H1. Network Topology Hypothesis -The dissemination of knowledge about Edubba in the citation network has a hierarchical (tree-like) structure: primary sources are the roots, interpretive and authoritative works are the branches?

H2.Structural Gap Hypothesis - Primary archaeological sources(step1) are structurally isolated from the most cited authoritative works(step3), there are no direct or indirect citational links between them?

H3.Bottleneck Hypothesis - The only structural bridge between the primary sources (Step 1) and the authoritative literature (Step 3) is an extremely small layer of mediators (Step 2), which forms a bottleneck. This effect is a structural property of the network, not an artifact of the classification method.

H4.Epistemic Displacement Hypothesis - Structural influence in the network (eigenvector centrality) is concentrated in Step 3, not Step 1 primary sources are rarely involved in dense, interconnected clusters. Significance is formed not where the data originates, but where interpretations are repeatedly cited?

H5.Epistemic Reliability Rarity Hypothesis - Sources that simultaneously satisfy all three criteria of trustworthiness (connection to primary data, high in-degree, and mutual reinforcement via the Trust Triangle) are structurally rare (less 1% of the corpus) and belong to mediators (Step 2) rather than authorities (Step 3)?

Data

Sources were selected based on two criteria: exclusive academic affiliation with the topic of scribal education in the Old Babylonian period, and unique DOI identifiers ensuring verifiability. Large monographs covering heterogeneous topics were excluded, with exceptions for works focused exclusively on Mesopotamian scribal practices. In parallel, archaeological excavation reports were collected as primary sources (Step 1) and stored in a structured CSV

¹**Step-Distance Model** a five-level classification of sources in which:

Step 1 denotes direct primary sources (archaeological publications providing empirical data)

Step 2 mediators (interpretive works citing Step 1)

Step 3 authority sources (canonical works with high in-degree, cited by Step 2)

Step 4 mediated sources (synthetic works without direct primary grounding)

Step 5 peripheral sources (all remaining works)

More detailed description can be found in the github repository ([./7-documentation/](#))

[docStep-Distance Model.md](#)

² More detailed description can be found in the github repository ([./7-documentation/](#))

[docDataAbstraction.md](#)

file (sourcesArchaeologicalRelies.csv) including site information, excavation metadata, tablet counts, and references. These were entered manually. All data collection process were explained in github repository³

Data Abstraction

The dataset is a directed network (graph) in which nodes represent individual academic publications and directed edges represent citation relationships. This is a relational dataset of type items + links, where the analytical interest lies not in the attributes of individual items in isolation, but in the topology of their connections. Analysis includes Step-Distance Classification. Each source was assigned a ‘step distance’ value (1–5)⁴. The step-distance model shifts the analytical frame from popularity (how many times a source is cited) to epistemic proximity (how close a source is to verifiable primary data). This distinction is the analytical core of RQ2 and hypotheses H2–H5: it allows the study to ask not just which sources are central, but where in the knowledge chain centrality emerges. The directed citation graph was constructed using igraph (Python).

Full corpus: 2,103 nodes, 2,524 edges (density = 0.0006).

Epistemic subgraph (Steps 1–3, top 15% betweenness filter): 400 nodes, 841 edges (density = 0.0053).

Methods

The analytical workflow is structured as a sequential hypothesis-driven investigation. The entry point is the global structure of the network (H1). Each subsequent hypothesis zooms into a specific structural property revealed by the preceding finding.

All figures use the unified design system (defined in design_system.py) and use the DejaVu Sans font and a muted colour palette. The visual design includes a white background, removed top and right spines, and semi-transparent annotation boxes with alpha transparency⁵.

H1 The dissemination of knowledge about Edubba in the citation network has a hierarchical (tree-like) structure: primary sources are the roots, interpretive and authoritative works are the branches?

Description: If the structure is hierarchical, the distribution should be balanced between steps from the primary source to peripheral sources (each level is comparable in the number of nodes).

Method: Step-Distance Model is a five-level classification of each node based on a combination of network metrics. Each source is assigned a step_distance

Steps	Classification Criteria
Step 1	Direct links to primary archaeological data (completed manually)

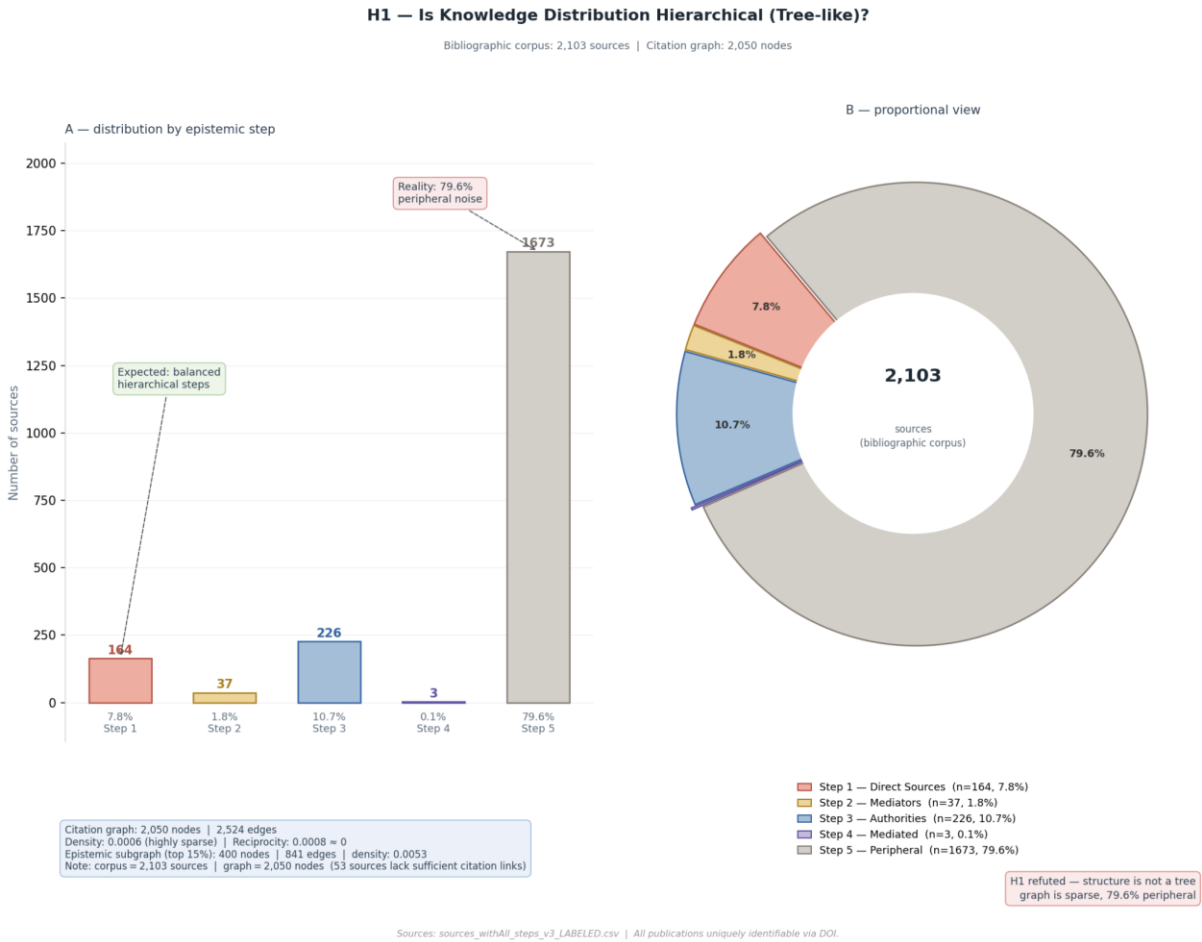
³ More detailed description can be found in the github repository [docDataCollection](#) (./7-documentation/)

⁴ More detailed description can be found in the github repository [docDataAbstraction.md](#) (./7-documentation/)

⁵ Alpha transparency refers to the use of an alpha channel to control the opacity of graphical elements, allowing partial visibility of underlying layers (Porter & Duff, 1984)

Step 2	Betweenness $\geq$ top 15% AND cites $\geq$ 1 node Step 1
Step 3	In-degree $\geq$ top 10% AND cites $\geq$ 2 nodes Step 2 AND age $\geq$ 10 years
Step 4	Does not cite Step 1 AND cites $\geq$ 2 Step 3 AND out-degree $\geq$ 2
Step 5	All others

Visualization:



Data: sources_withAll_steps_v2_LABELED.csv (N=2050) + citation_graph.csv

Figure 1. Global Graph: Step Distance Distribution (full graph, N = 2,103).

Type of Diagram

Type A - Bar Chart (C + Q): allows for precise comparison of the number of nodes at each step. The x-axis is categorical (step), the y-axis is quantitative (count). According to the precision hierarchy, position along a common scale is the most precise channel for quantitative comparison⁶.

⁶ Cleveland & McGill, 1984

Type B Donut Chart (part of whole): shows proportions. A donut was chosen instead of a pie because the central area allows for the total number of sources to be shown without loss of readability. Annotations show expectation (green) and reality (red).

Why this Visualisation Method not other: A bar chart was chosen for Panel A because the task is Compare comparing discrete categories (5 steps) on a single quantitative metric (count). According to Cleveland & McGill (1984), position along a common scale is the most accurate perceptual channel for magnitude comparison, superior to area (pie chart) or angle. A pie chart was explicitly avoided because it forces the viewer to compare arc lengths and angles channels with known perceptual limitations for precise comparison.

A donut was chosen for Panel B over a full pie because the central hole allows the total N to be annotated without cluttering the segments. The donut form also reduces the area of the largest segment (Step 5), making proportional differences more readable.

How to read: Read Panel A left to right: Step 1 through Step 5. A hierarchical structure would produce a roughly bell-shaped or pyramid distribution comparable to bars for Steps 1–4 with Step 5 smaller. Instead, the bar for Step 5 dominates visually ($n = 1,673$, 79.6%), while Steps 2 and 4 are barely visible. Panel B reinforces this: the near-total grey segment of Step 5 makes the imbalance immediately apparent. The green annotation marks what would be expected in a hierarchy, the red annotation marks the actual result.

Result: **H1 Not Confirmed.**

The graph is not a tree, not embedded in any hierarchical chain. The graph is extremely sparse (density=0.0006, reciprocity=0). The structure is not tree-like 79.6% of nodes are peripheral and not embedded in any hierarchy. The graph has a scattered structure with several dense clusters rather than a tree.

Distance	Source Role	Source Function	Citation Network Method	Edge Quantity	Percentage distribution
Step_1	Direct sources	Publications providing direct empirical evidence — excavation reports, tablet publications, and site documentation	Filled Manually	164	7.8%
Step 2	Mediators	Interpretive works that cite primary data and transmit empirical findings into the broader scholarly discourse structural bridges between evidence and interpretation"	betweenness $\in$ top 15% AND cites $\geq$ 1 Step1 nodes	37	1.8%
Step 3	Authorities (High in-degree)	Canonical sources within this corpus works whose interpretations have stabilised and are repeatedly cited as authoritative reference points	in-degree $\in$ top 10% AND cited by $\geq$ 2 Step2 nodes AND publication age $\geq$ threshold (10 years)	226	10.7%
Step 4	Mediate d source	Synthetic works - comparative studies and reviews that consolidate authority-level interpretations without direct grounding in primary evidence	cites no Step1 AND cites $\geq$ 2 Step3 AND out-degree $\geq$ 2	3	0.1%

Step_5	Peripheral	All others	NOT direct reliance NOT Mediators NOT Authority NOT Mediated Sources	1673	79.6%
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Table 1. Step-Distance Model: epistemic role assignment criteria and corpus distribution (full graph, N = 2,103).

Step 5 absorbs both truly peripheral works and artifacts of corpus incompleteness . This led to the decision to extract a subgraph (top 15% betweenness) for more detailed analysis (N=400, 841 edges, density=0.0053).

This result raises the following question: If the structure is not hierarchical, where are the primary sources in relation to the authoritative ones (Step 3) or do they exist in isolated areas of the network?

H2 - Primary archaeological sources(Step 1) are structurally isolated from the most cited authoritative works (Step 3), there are no direct or indirect citational links between them

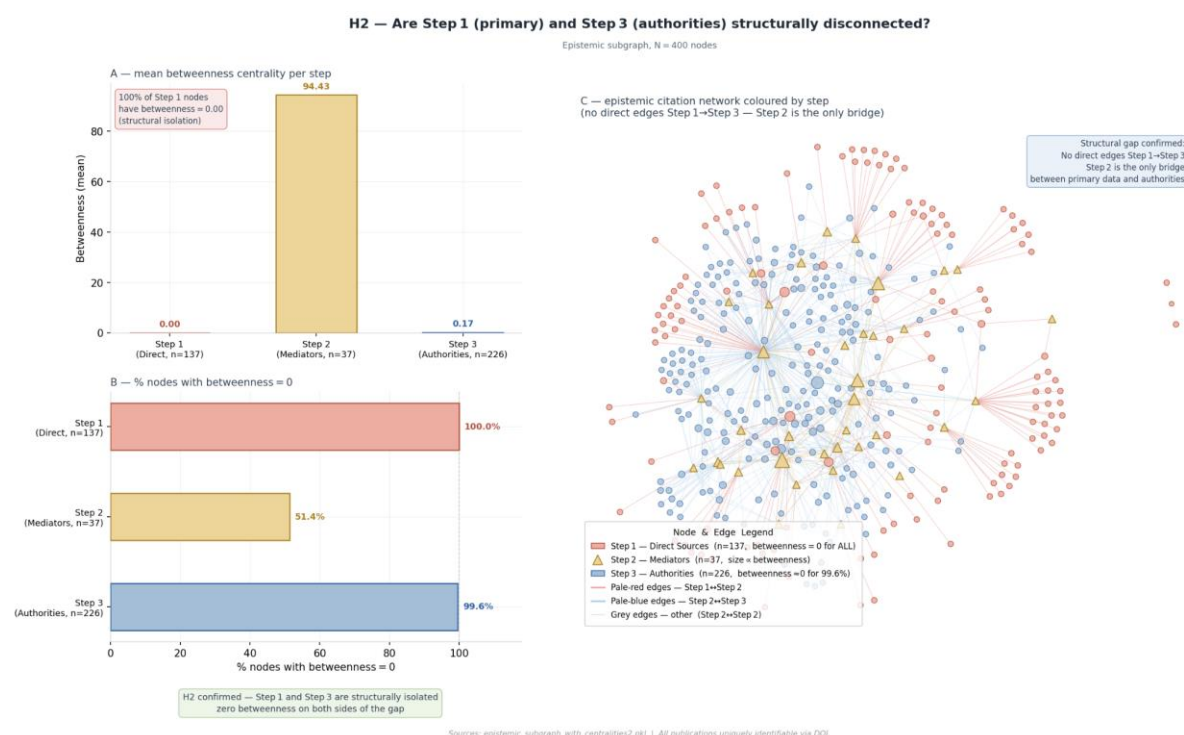
Description: After discovering the sparseness, it was checked whether there was any connection between Step 1 and Step 3.

Method:

To test whether Step 1 and Step 3 are structurally disconnected, betweenness centrality⁷ was used as the validation metric: a node with betweenness = 0 lies on no shortest path between any other pair of nodes, meaning it neither transmits nor mediates information flow within the network. Betweenness was computed for all nodes in the epistemic subgraph (N = 400) and compared across steps to determine whether primary sources participate in any citation pathway connecting them to authoritative works.

Visualization:

⁷ Betweenness centrality measures the extent to which a node lies on shortest paths between other nodes and therefore occupies a potentially important intermediary position in the network . (Easley, Kleinberg, 2010)



Data: epistemic_communities2.csv — epistemic subgraph, N=400 (Steps 1–3)

Figure 2. Betweenness Centrality

Type of Diagram:

Panel A Bar Chart (O + Q): Comparison of mean betweenness across steps. Step 2 (mean=94.43) contrasts with Step 1 (0.00) and Step 3 (0.17). The bar chart was chosen because three discrete categories need to be compared using a single quantitative metric, this is a Query - Compare task, solved using position along the common scale (the most accurate channel)⁸.

Panel B Horizontal Bar Chart (proportion of nodes with betweenness=0): Shows that 100% of Step 1 and 99.6% of Step 3 are not involved in information transfer. The horizontal orientation was chosen to show the progression from 0% to 100% and make the comparison of proportions clear.

Panel C: Panel C uses a node-link graph rather than a matrix because the dataset is sparse (density = 0.0053 in the subgraph) and the topological pattern, the absence of edges between Step 1 nodes and Step 3 nodes is the visual finding itself. A matrix would make this absence harder to perceive. In a node-link layout, the spatial gap(no connection) between red and blue node clusters is immediately visible.

How to read: In Panel A compare bar heights across the three steps. Step 2 (amber bar) should be visibly taller. In Panel B the near-total fill for Steps 1 and 3 (100% and 99.6% betweenness = 0) versus Step 2 confirms structural isolation. In Panel C Observe the spatial separation of the pale terracotta (red) cluster of Step 1 nodes and the pale blue cluster of Step 3 nodes. Pale amber triangles (Step 2) act as the only visible bridge.

Result: **H2 Confirmed.**

⁸ Cleveland & McGill 1984

All 137 Step 1 nodes have betweenness = 0.00. No primary source lies on the shortest path between any two other nodes in the network. Primary data are dead ends, not transit nodes.

Step 3 also participates virtually in no information transfer (99.6% betweenness = 0). There are no direct connections from Step 1 to Step 3.

Step 1 has a non-zero in-degree (mean=1.33, max=9) they are cited, but they themselves do not serve as transit points.

Note: Step 1 is not invisible mean in-degree = 1.33 (max = 9 for Gasche1989). Primary sources are cited, but information does not pass through them further down the network.

This result raises the following question: If Step 1 and Step 3 aren't directly linked, who is the intermediary? Is there even a path from primary sources to authority?

H3 - The only structural bridge between the primary sources and the authoritative literature is an extremely small layer of mediators, which forms a bottleneck. This effect is a structural property of the network, not an artifact of the classification method

Description: H2 establishes that Step 1 and Step 3 are isolated. But isolation doesn't mean there's no path at all, it could mean the path must go through a third party. H3 checks who this third party is, how stable it is, and whether it's a random consequence of the chosen filtering threshold.

Method 1 - Analyzing the betweenness distribution by Step 2. If Step 2 is indeed the only bridge, then its nodes should have the highest betweenness centrality, that is, they should pass through the largest number of shortest paths in the network.

Step 2 (n=37) - single layer with high betweenness

Robson2001	Betweenness	802.17,	In-degree	9
Cammarrano2014	Betweenness	551.17,	In-degree	3
Pearce1995	Betweenness	541.25,	In-degree	3
Charpin2010	Betweenness	421.50,	In-degree	2
George0000	Betweenness	376.25,	In-degree	14

Method 2 - Sensitivity Analysis.

What if ?

Criterion A: Betweenness $\geq$ top15%

What if ?

Criterion B: Quotes $\geq$ 1 node Step 1

What if criterion A is artificial? What if I chose the "top 15%" threshold arbitrarily, and it's this threshold that's creating the bottleneck—not the actual network structure?

If we remove the betweenness filter, will the number of mediators change? If not, the bottleneck is independent of the threshold and is a real structural property.

Sensitivity analysis:

with betweenness filter = 37 nodes; without filter = 37 nodes (unchanged)

Step1 loaded: 137 nodes
Computing centralities...
Step2 (mediators): 37
Step2 (mediators, no_betweenness filter): 37
Step3 (authorities): 226
Step4 (mediated): 3

Result: Confirmed.

Step 2 is the only bridge between Step 1 and Step 3. Only 37 nodes (1.8%) have connections to primary sources. The bottleneck is a structural property of the network, and used as the sources for interpretations. The concentration of betweenness is critically high in 5 key mediators.

Visualization:

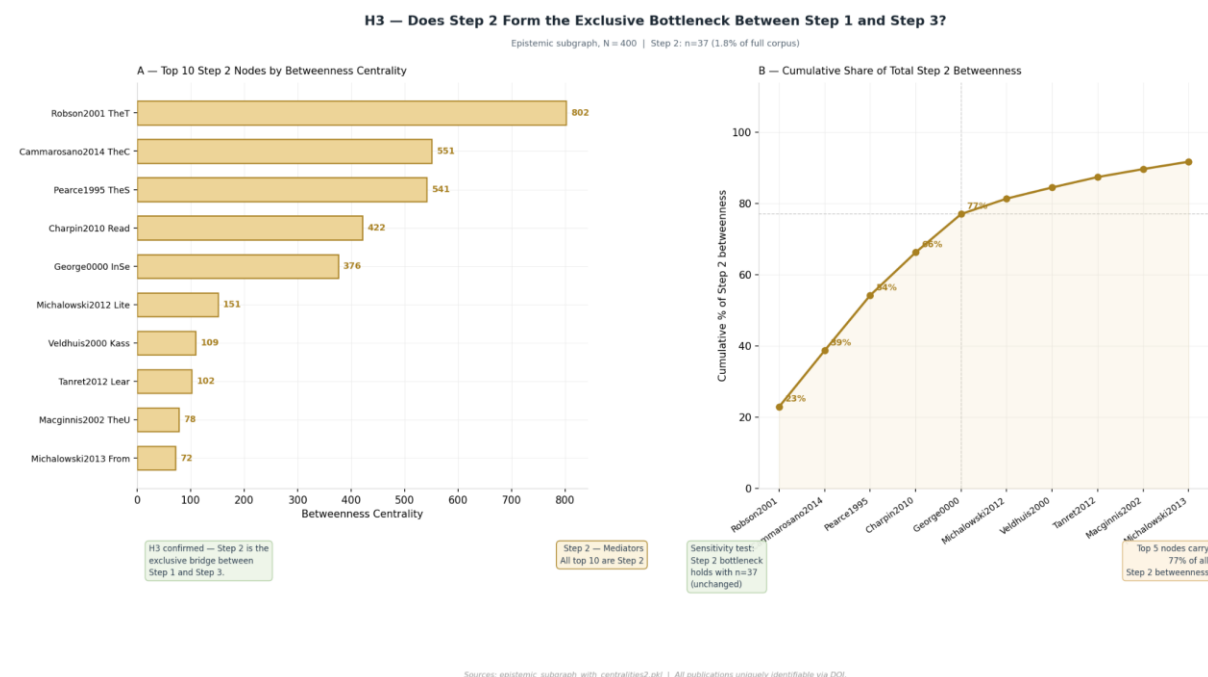


Figure 3. High Betweenness

Type of Diagram:

Panel A Horizontal Bar Chart (C + Q, ranked). Top 10 nodes by betweenness. The horizontal orientation was chosen because the task is Query - Identify extremes, we need to show the ranking and allow the reader to quickly identify the leaders. Horizontal bars are better than vertical ones for long labels (source_id). All 10 nodes are Step 2 with top 10 highest betweenness, which immediately visually confirms the hypothesis through the color hue (a single amber color).

Panel B Line Chart (cumulative share): Shows how concentrated the betweenness is within Step 2. The cumulative curve is a standard method for analyzing concentration (similar to the

Lorenz curve)⁹. The line chart was chosen because the x-axis represents ordinal positions (ordered categorical), and a line chart conveys accumulation better than individual bars. Query-Summarize.

Why this Visualisation Method not other: Panel A uses a horizontal bar chart with ranked ordering because the task is Identify extremes showing which Step 2 nodes dominate and enabling the viewer to read long source identifiers. A vertical bar chart would require diagonal labels. A scatter plot was considered but rejected: it distributes points along two axes and makes rank comparison harder than aligned bars on a common scale. Panel B uses a line chart rather than a histogram because the x-axis represents ordinal positions (ranked nodes) and a line conveys accumulation the idea that the betweenness "fills up" as we move through the top mediators more naturally than individual bars. This is analogous to a Lorenz concentration curve: the steeper the curve, the more concentrated the bottleneck. The task here is to summarize distribution shape, not compare individual values.

This result raises the following question: If Step 2 is the only bridge, how is the structural influence distributed(eigenvector) to sources close to the primary data or to sources far from it?

H4 - Structural influence in the network (eigenvector centrality¹⁰) is concentrated in Step 3, not Step 1 primary sources are rarely involved in dense, interconnected clusters. Significance is formed not where the data originates, but where interpretations are repeatedly cited.

Description: After discovering the bottleneck, it was checked where the "network influence" was concentrated.

Method: A node has a high eigenvector not simply because it is widely cited, but because it is cited by sources that are themselves widely cited. This is a measure of its embeddedness in dense, mutually reinforcing clusters -something that corresponds to "discursive authority" in a knowledge network.

If H4 is true, the mean eigenvector of Step 3 should be significantly higher than that of Step 1. Step 1 should have a high proportion of nodes with eigenvector = 0.

Result Confirmed with an important exception:

-80.3% of Step 1 nodes have eigenvector = 0. The average Step 3 (0.103) is higher than Step 1 (0.059).

-Anomaly of Kramer's (1949). The only Step 1 node with a maximum eigenvector = 1.0.

"Schooldays" by Kramer is one of the first descriptive texts about Edubba, cited by 7(in-degree) highly central Step 3 sources in community 5. It is "pulled" toward the core via incoming citations, but its betweenness = 0.0 means it is a dead end, not a bridge.

-Step 2 is a paradox. Mediators have the smallest mean eigenvector (0.035) of all three steps even lower than Step 1. Step 2 is a bridge in betweenness, but not in eigenvector. This means

⁹ The Lorenz curve is a graphical tool used to represent the cumulative distribution and concentration of a variable across a population (Cowell, 2011).

¹⁰ Eigenvector centrality is a measure of node importance in which a node receives a high score if it is connected to other nodes that are themselves highly central. (Easley, Kleinberg, 2010)

that mediators(Step 2) are not structurally embedded in dense clusters, they connect isolated zones without being part of the network's elite.

Eigenvector Centrality

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step_distance = 1 | n = 137 | mean eig = 0.05944 | moderately significant
step_distance = 2 | n = 37 | mean eig = 0.03475 | structurally weak
step_distance = 3 | n = 226 | mean eig = 0.10303 | concentration centrality

```

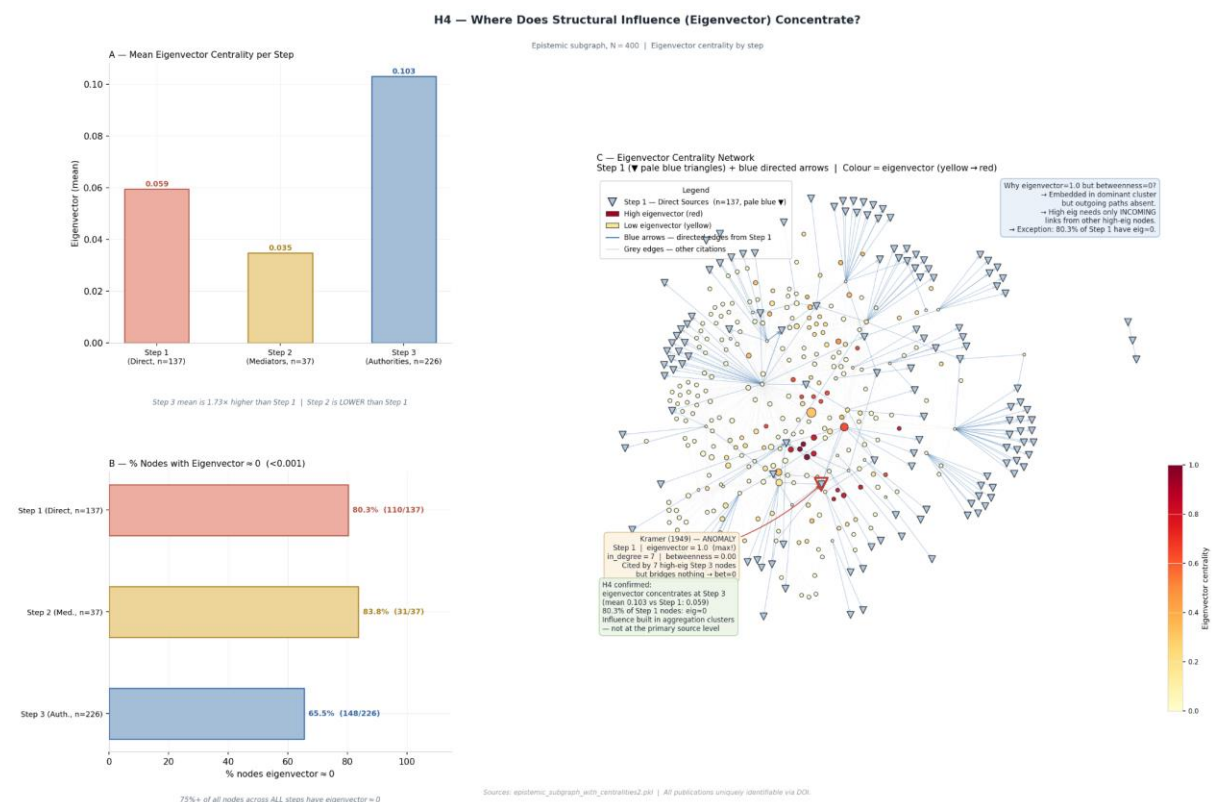
Eigenvector Centrality:

```

step_distance = 1 | median: 1.625 | max: 1.0
step_distance = 2 | median: 0.0 | max: 0.571
step_distance = 3 | median: 2.581 | max: 0.857

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Visualization:



Data: epistemic_communities2.csv — epistemic subgraph, N=400

Figure 4. Eigenvector Centrality

Type of Diagram

Panel A Grouped Bar Chart demonstrates mean eigenvector per step. **Panel B** Horizontal Bar Chart shows percentage of nodes with eigenvector = 0. **Panel C** Network Heatmap demonstrates the same node-link layout as in H2, with nodes colored by a YlOrRd¹¹ color scale encoding eigenvector value. Step 1 nodes rendered as blue triangles.

¹¹ Munzner T.(ch.10,2010)

Why This Visualisation Method, Not Another:

Panels A and B use bar charts for the same reason as in H2. The task is compare across discrete step categories. Panel C uses a network heatmap, a node-link graph with a sequential color scale applied to nodes (from red nodes with high eigenvectors and yellow with low) rather than a standalone color-encoded scatter plot. Sequential color scale *YlOrRd* (“Yellow–Orange–Red”) from the ColorBrewer palette system was used to encode increasing quantitative intensity. Sequential palettes are recommended for ordered numerical data because gradual luminance and hue variation support perceptual ranking and magnitude comparison.

The network layout preserves topological context (which nodes are connected to which), allowing the viewer to observe that high-eigenvector nodes cluster together spatially and are positioned at Step 3, while Step 1 triangles (blue) sit isolated at the periphery. A scatter plot of eigenvector vs. in-degree would show correlation but would lose the spatial/topological dimension

How to read: In Panel C Network Heatmap warm-colored (yellow/red) nodes represent high-eigenvector publications, which demonstrate that most popular sources (red nodes) are not connected to primary sources (blue triangle). The blue triangles consistently appear in cool colors (yellow/pale) at the periphery. The one exception is Kramer1949, annotated explicitly.

This result raises the following question: If neither closeness to the primary data (Step 1) nor a high eigenvector (Step 3) alone guarantees reliability, can we find sources that meet all three conditions together?

H5 - Sources that satisfy all three criteria of trustworthiness at once connection to primary data, high in-degree, and reinforcement via the Trust Triangle-are structurally rare (less than 1% of the corpus) and are classified as mediators (Step 2) rather than authorities (Step 3)

Description: The final step is to apply the Trust Triangle as a composite reliability criterion.

Method Trust Triangle:

A source is reliable if three conditions are simultaneously met:

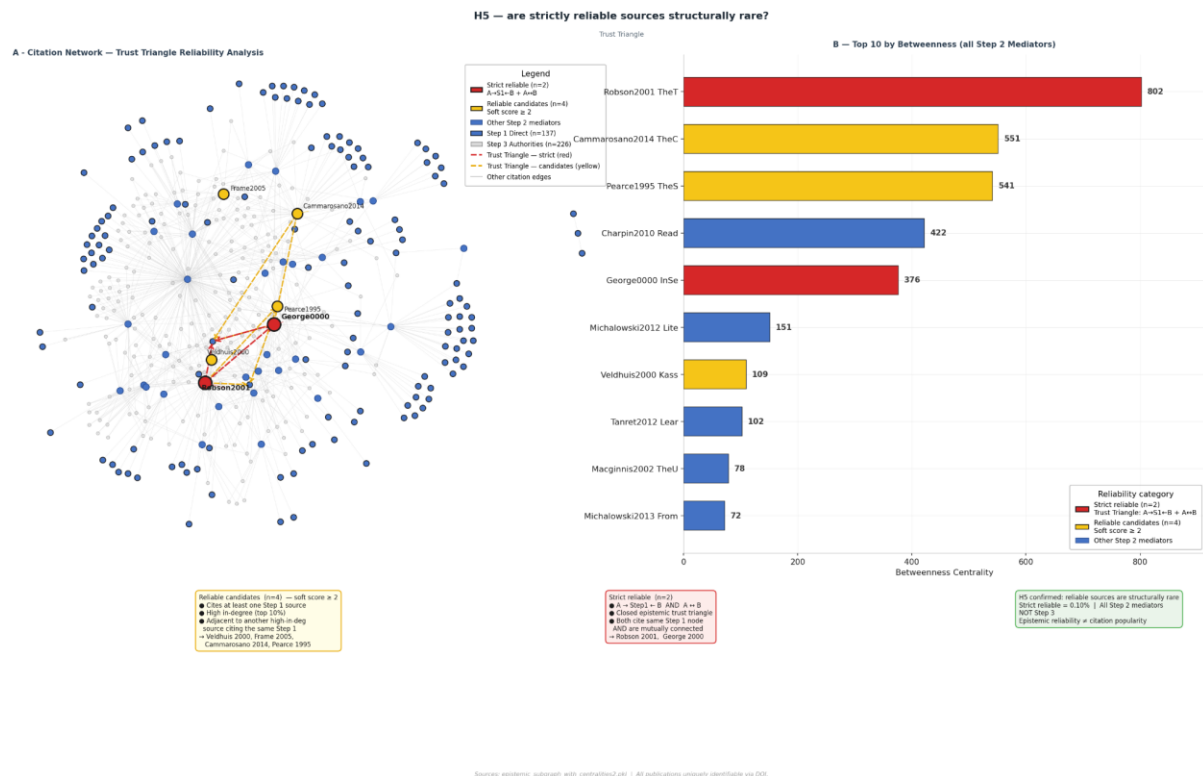
Condition 1 Link to primary data. Node A directly cites at least one Step 1 source.

Condition 2 High in-degree. Node A has a high in-degree (is widely cited by other sources).

Condition 3 Mutual reinforcement. There is another node B with a high in-degree that cites the same Step 1 source as A, forming a closed trust triangle.

This triangle distinguishes trustworthiness from random citation: it's not just "someone cited the primary source," but "several independent authoritative works have confirmed the same fact."

Visualization:



Data: sources_withAll_steps_v2_LABELED.csv (full dataset, N=2,103) + epistemic_communities2.csv (subgraph, N=400)

Figure 5. Trust Triangle -Reliable Sources.

Type of Diagram:

Panel A - Trust Triangle Citation Network (node-link graph, same layout as H2/H4) with two reliability tiers rendered as dashed triangular edges.

Panel B - Betweenness bar chart showing that all reliable sources are Step 2, not Step 3 or others.

Why this Visualisation Method not other: The node-link graph was chosen because the trust triangle is a topological property: a closed triad of three nodes and three edges. It cannot be represented by a bar chart or scatter plot without losing the relational structure. Showing the actual triangle edges (as dashed lines) makes the geometric definition of the criterion visually self-explanatory: the viewer sees exactly what "closed triangular configuration" means without reading a definition. Consistent node positions allow the reader to recognise familiar clusters and locate reliable sources relative to the bottleneck structure already shown. The use of dashed edges for trust triangle links is a deliberate choice, the viewer perceives the three dashed edges as a triangle even when other edges are present, because the dashed style pre-attentively groups them into a distinct visual object.

How to read: Strong reliable sources - two red nodes (Robson2001 and George0000), the strictly reliable sources. Three dashed red edges connect them to each other and to a shared Step 1 (blue) node, forming a visible triangle. Yellow nodes (top-4 soft candidates) are nearby but do not complete a fully closed triangle. The vast majority of nodes are grey or blue, with no triangular configuration, confirming that reliability is structurally rare.

Result: Confirmed with an unexpected conclusion.

Reliability is not the same as popularity. The most epistemically reliable sources are not the most cited ones, but those that stand at the intersection of primary data and authoritative clusters. By definition Step 3,¹² it does not cite Step 1, its betweenness = 0 (99.6% of nodes). Without a connection to the primary data, condition 1 is not satisfied. For example, the most cited source in Step 3 (Adams 1969, in-degree=21) failed the Trust Triangle: betweenness=0, no direct connection to the primary data. Citation popularity and epistemic reliability are different structural properties.

Reliable sources aren't just scarce, they structurally belong to a different layer than expected. Authorities (Step 3), the discipline's most well-known layer, failed the reliability test. Mediators (Step 2), a little-known intermediate layer, were the only candidates satisfying all criteria simultaneously.

Discussion

A central contribution of this study lies in the visual operationalisation of epistemic relations within scholarly discourse. By transforming citation relationships into a multi-layered visual system, the analysis makes visible structural properties that remain inaccessible through traditional close reading or bibliographic review.

Across all five figures, a consistent design system was employed, enabling cross-figure visual comparison of epistemic roles. This consistency is methodologically significant: it allows the viewer to interpret Step 1–Step 5 not as isolated analytical categories, but as a continuous spatial and visual encoding of epistemic distance from primary data.

Figure 1 demonstrates that the citation network does not approximate a hierarchical structure. The dominance of Step 5 nodes, clearly visible in both bar and donut representations, immediately contradicts the expected tree-like model. Importantly, this conclusion does not depend on numerical output alone. The redundancy of visual encodings (bar length + proportional area) ensures that the absence of hierarchy is perceptually robust: even without reading values, the structure is immediately identifiable as non-hierarchical.

This highlights a broader methodological point: hierarchical epistemic models may persist in theory, but fail under visual structural inspection of citation systems.

Figure 2 reveals one of the most important structural findings of the study: primary archaeological sources (Step 1) are not integrated into the flow of scholarly mediation.

The node-link visualization is crucial here. Unlike tabular metrics, the spatial separation between Step 1 and Step 3 clusters makes the structural disconnection immediately perceptible. The absence of connecting edges is itself a visual result.

Figure 3 demonstrates that epistemic flow is highly concentrated within a small subset of Step 2 nodes. The Lorenz-like accumulation curve in Panel B is particularly important: it visually encodes extreme inequality in betweenness distribution. This is a key methodological strength of the visualization: it transforms abstract centrality concentration into an immediately interpretable geometric pattern. The result is not merely quantitative but structural epistemic connectivity is not distributed, it is compressed into a minimal bridging layer. The visual design

¹² Step 3 is defined as nodes with an in-degree $\geq$ top 10% AND cited by $\geq$ 2 Step 2 nodes AND age $\geq$ 10 years.

(uniform color for Step 2 nodes) further reinforces this interpretation by removing categorical noise and emphasising internal rank only.

Figure 4 reveals a counterintuitive pattern, eigenvector centrality is not aligned with proximity to primary data. The heatmap representation is crucial here, because it preserves both network topology (who connects to whom) and scalar intensity (eigenvector magnitude).

This dual encoding makes visible a structural mismatch:

- Step 1 nodes occupy peripheral, low-intensity regions and
- Step 3 nodes dominate high-density eigenvector clusters.

The anomaly (Kramer 1949) is particularly important because it is visually isolated. It appears as a high-intensity node in a structurally low-centrality class. This makes it an epistemic outlier rather than a category contradiction. Thus, visual analysis supports the conclusion that epistemic influence is decoupled from empirical proximity.

Figure 5 provides the most complex visual argument of the study. The Trust Triangle is inherently a topological object, and therefore requires a network-based visualization rather than a scalar representation. The key methodological point is that reliability is defined not by individual attributes, but by closed relational configurations. This makes the visualization indispensable, the concept cannot be understood without seeing triangular closure. Reliable sources are extremely rare (visually sparse red nodes). They cluster in Step 2 rather than Step 3. This is particularly significant because it directly contradicts the intuitive expectation that authority (Step 3) corresponds to reliability. Instead, the visualization shows that reliability emerges from structural mediation patterns, not citation popularity.

Limitations

The node-link diagram is useful for small subgraphs but does not scale to the full dataset, because visual complexity increases faster than human perceptual capacity. Therefore, alternative representations are required for global structure analysis.

The step-distance model is based on citation structure rather than content analysis. It therefore assigns epistemic roles according to network position, not the correctness of individual claims. As a result, the visualisations should be read as representations of knowledge structure, not as indicators of scholarly validity. These figures capture relational patterns in the citation network, not semantic or qualitative evaluation of the sources.

Step 4 is under-represented in the dataset, with only three identified sources. This is likely due to the DOI-based sampling strategy, which favours journal articles over textbooks and encyclopedic works. As a result, the visualisations may under-represent synthesis-level scholarship. The Step 4 layer in the figures should therefore be interpreted as a partial reflection of the epistemic landscape rather than a complete one.

Conclusion

This project maps the architecture of Edubba scholarship using directed citation network analysis and information visualisation, applying the analytical framework developed throughout this course. A key outcome of this research is methodological rather than purely substantive. The use of coordinated network visualisations demonstrates that epistemic structures in scholarly fields are not readily observable through linear reading or citation counts alone. Instead, they become intelligible through multi-layered visual encodings that combine topology, centrality, and categorical role assignment. The consistent visual grammar applied across figures allowed structural comparisons that would not be possible through isolated statistical outputs.

Importantly, the findings challenge a common assumption in scholarly communication: that citation frequency and epistemic reliability are aligned. In the case of Edubba scholarship, these dimensions diverge. Highly cited works are not necessarily those most structurally connected to primary data, and reliable sources emerge from a narrow and structurally specific position within the network.

At the same time, the study highlights the importance of intermediaries in knowledge production. Rather than functioning as passive connectors, mediating sources constitute the core structural mechanism through which archaeological evidence is transformed into scholarly interpretation. In this sense, the epistemic system is not hierarchical but translation-driven, with mediation playing a central structural role.

Finally, this project demonstrates the value of applying information visualisation techniques to humanities datasets not merely as illustration, but as an analytical instrument. By making invisible relational structures visible, visual analytics enables a shift from interpretive description to structural explanation.

Future work could extend this approach by incorporating temporal dynamics of citation formation, testing the robustness of the Trust Triangle under alternative definitions of reliability, and comparing the Edubba network with other archaeological or historical domains to assess whether the observed mediator-dominated structure is domain-specific or generalisable.

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Software and Digital Tools

- **Python** — data processing and analysis (<https://www.python.org>)
- **igraph** — network construction and analysis (<https://igraph.org>)
- **AnyStyle** — bibliographic reference extraction (<https://anystyle.io>)
- **pdfplumber** — PDF text extraction (<https://github.com/jsvine/pdfplumber>)
- **pytesseract** — OCR for scanned documents(<https://github.com/madmaze/pytesseract>)
- **hslpicker** - color analyses and design (https://hslpicker.com/?utm_source=chatgpt.com#efb1a6)

Вкладка 2

Что реально интересного обнаружено в данных

1. Надёжные источники — это медиаторы, не авторитеты: Все 4 источника с `reliable_source=1` — Step 2:

- Robson (2001), George (год неизвестен), Veldhuis (2000), Frame (2005) Это **противоречит интуитивному ожиданию** — самые надёжные структурно не самые цитируемые.

2. Аномалия Step 1 в eigenvector: Kramer (1949) — Step 1, но eigenvector = **1.0** (максимум!). Это исключение полностью опровергает H1 в сильной формулировке и требует отдельного объяснения в репорте.

3. Betweenness — почти исключительно Step 2: Топ-10 по betweenness — 9 из 10 узлов это Step 2. Robson2001 — betweenness=802, George0000 — 376. Это подтверждает H4 количественно и очень убедительно.

4. Сообщества — почти все изолированные: 203 сообщества, из которых **одно** (community=5) содержит **184 узла**, а остальные 202 сообщества содержат 1–3 узла. Это говорит об экстремальной фрагментации сети — важная находка.

5. Расхождение Step1: 164 vs. 137 — 27 первичных источников есть в полном датасете, но не вошли в подграф. Это стоит упомянуть как limitation.

6. 96% источников — "не primary text" поле `is_primary_text` практически не заполнено. Это проблема данных, которую надо честно ук